Nguyễn Thiện Nhân - 2452879 - HW5.

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TEKO SOLUTION

1.

For 12 candidate features, I use chromosome of 12 bits, each one represent 1 feature. The chromoseme is decoded by collecting indices of all bits equal to 1 and and use those features to train predictive model. An undesirable chromesome is all 0, I would handle by rejecting it and regenerating empty chromosomes or repairing them by mutation.

2.

I define fitness function: $F = Accuracy - 0.01k$, where k is number of selected feature. For chromosome 3 feature, accuracy 0.83 $\rightarrow F = 0.83 - 0.01 \cdot 3 = 0.8$, chromosome 9 feature, accuracy 0.88 $\rightarrow F = 0.88 - 0,01 \cdot 9 = 0.79 \rightarrow$ I'd choose chromosome 3 feature because it achieve higher fitness. The fitness should balance between predictive performance and model simplicity and should be selected based on system requirements.

3.

Starting from 101001011100, apply mutation: 101101011110. Then I choose 2nd parent 110110100011, single point crossover after 6th bit. → off spring 101011000011. These changes work because individuals bits can be exchanged or flipped independently. However, ordinary

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crossover and mutation maybe inappropriate for permutation representation. So they require specalized operators like ordes crossover and swap mutations.

4.

A possible cause of premature convergence is exessive selection pressure, allowing high-fitness individuals dominates → eliminate exploration. Another cause is low mutation rate → limit genetic variation and hard to discover new regions of search space. To address first cause, I'll use lower selection pressue, allowing more candidates to reproduce. To solve the second cause., I'll increase mutation rate and introduce random immigrants to add new gentic materials.

5.

I use a GA for feature selection, with each solution = a binary chromosome. Fitness function reward accuracy and penalize feature selection. Population = 50, stop after 100 generations or 20 without improvements. I'd use 3 seeds and report best fitness and avg Hamming distance between chromosomes in each run. Multiple runs are necessary because genetic algorithms are stochastic., so 1 successful run doesn't mean reliability overall.